

Final Project — Submission Guidelines

What I want from you

1. **Code** — a working Python script or notebook implementing your method.
2. **Presentation** — a short write-up / slide deck covering motivation, method, validation results, and what did not work. Cross-validation is required wherever it applies; numbers without out-of-sample evaluation are not evidence.

Deadline: both the code and the presentation must be sent to me via email at least 48 hours before your presentation datetime.

Project 2

- **Trade-side classification.** Given trade prices and volumes, with no quotes, infer the aggressor side of each print. Beat or motivate beyond Lee–Ready / tick rule / quote rule baselines.

Code interface for the structured projects

For Project 2, wrap your method as a single class or function with a clean signature. I should be able to import it and run it on my own data.

Project 2

```
def classify_side(  
    trades: pd.DataFrame  
) -> pd.Series:
```

Input is a pandas DataFrame with trade price and volume columns, time-indexed. Output is a boolean pandas Series aligned to the input where **True** means sell aggressor and **False** means buy aggressor, matching the convention of the sample data.

Sample data

Exemplary crypto trades and order-book data is attached for you to develop and validate against. Conventions:

- Trades: the aggressor-side flag is boolean — **True** means sell aggressor, **False** means buy aggressor.
- Order books: standard L2 snapshots, with bids and asks including prices and sizes.